Khostty State & Forward Docket v1.0 @ 8e86ec93a

Khostty — State & Forward Docket v1.0

Product: KooshaPari/Khostty — a Phenotype-flavored fork of `ghostty-org/ghostty` **Snapshot:** `2026-10-17T22:17Z` **Subject revision:** `main` @ `8e86ec93a1fd0f0c18f39b5c2d9ad604c1ffbaca` **Spec:** `~/CodeProjects/docs/docs-5/adr/2026-10-17-state-and-forward-docket-v1.md` **Author:** VP-Engineering (jcode orchestrator) — this run **Priority inputs (all committed, referenced by SHA):** `docs/audits-archive/2026-10-17-final/` @ `869b9c75e` (`FINAL-AUDIT.md`, `FWD-PATH.md`, `state.md`, `stubs.md`, `usability.md`, `mux.md`), `docs/sessions/20261017-mux-design/06_Q1_DECISION_MATRIX.md` @ `5708d2ba2`, `docs/sessions/20261017-mux-design/07_PHASE1_RISKS.md` @ `72ba2148f`

Binding (operator 2026-10-15): FORK-ONLY. No `gh pr/issue/discussion create` or `gh api POST/PATCH/DELETE` against `ghostty-org/ghostty` or any other upstream repo. Reading, copying, tracking, `zig build` against upstream: fine. Push to `KooshaPari/*`: fine.

File status: This file lives in `docs/audits-archive/`, which is **tracked** (the working `docs/audits/` is untracked by AGENTS.md binding rule #10). This docket may be committed normally.

Evidence rule: every claim below is either (a) read from a committed revision cited inline, (b) observed live in this run with the command shown, or (c) explicitly marked UNKNOWN. Nothing is inferred from a branch name, README percentage, or prior report alone.

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§0. Two-minute cockpit

Khostty is a **working library-first 0.1.0** (Rust crate on crates.io, WASM on npm, a signed macOS `.app`, a Linux GTK `.deb`, a Windows installer) with a **complete but un-wired agent-IPC surface**. The one thing that stands between the current tree and "the fork does what it was for" is **calling `Server.bind` from a running app runtime**. That is

blocked on a GTK-on-Linux host the operator has not yet provided, and on decision Q1 (which runtime Phase 1 targets).

What changed in this run (since the 05:18Z snapshot):

Change	Detail
Prettier drift cleared on 8 PR branches	57 tracked files were drifting; every PR touching them was red on a pre-existing failure
New PR #50	`chore/prettier-sweep-20261017` — systemic fix for `main`'s formatting drift
Session evidence committed	`b6977ebca` — the 07:50Z STOP report + phenoreg-linker report
Docket v1.0 published	this file + the framework spec

Progress quadrants (this snapshot)

```
RAW IMPLEMENTATION        ▓▓▓
```

Top three risks

Rank	Risk	Why it matters now
R1	**Agent-IPC runtime never wired** (`Server.bind` called by no runtime)	The fork's headline capability is unreachable until a GTK/Linux host exists. All 7,230 LOC of IPC is dormant.
R2	**Q1 undecided** (Phase 1 runtime: Linux+GTK / embedded-on-macOS / mock-only)	Blocks Phase D entirely; the mux-design docs recommend option (c) mock-only but the operator has not picked.
R3	**Windows GUI never launched**	The only runtime with a shipped installer that has never opened a window; S0-S8 track it; #30 (draft) is the vehicle.

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§1. Intent and boundary

1.1 Mature intended product

The parent outcome, stated in the PhenoRegistry dossier (`products/Khostty/DOSSIER.md`): **"a usable terminal or embedded terminal that maintains the required semantics of its upstream."** Concretely, for the fork to be "complete":

1. A person can install and run a terminal on their platform of choice and use it as a terminal.
2. An agent or embedding application can drive it over the fork's IPC surface.
3. Terminal correctness matches upstream on the conformance corpus.

1.2 Closest Viable Product (CVP)

Library-first 0.1.0 — the smallest identity-preserving slice that delivers a real outcome to a consumer:

Interface	Command	Outcome
Rust	`cargo add khostty-vt`	an instant VT parser in a Rust program
WASM/ JS	`npm install @khostty/libghostty-vt-wasm`	an instant VT parser in a browser/ Node program
C	link `libghostty-vt`	upstream's 203-function ABI, fork-built

CVP status: SHIPPED. `v0.1.0` tag = `7e1684f60e2ab03cca3e0b653c65b1b4e1247d10` (annotated, pointing at `202aad543`); GitHub release published 2026-09-27 with 6 assets; crates.io published 2026-09-20; npm published.

1.3 Minimum Viable Product (MVP)

CVP + the agent-IPC runtime bound on at least one platform. When `khostty` boots and an agent can `khostty-client ping` → `pong` and `pane.list` against a live instance, the fork delivers its differentiated capability. **MVP status: NOT DELIVERED.** `server.zig` exists (1,238 lines) but `Server.bind` is called from no runtime (audit-archive `state.md` §3.4; re-verified this run: the only `Server.bind` references are inside `src/apprt/ipc/`).

1.4 Exclusion list (what "done" does not include)

Per `FWD-PATH.md` §1 and the operator's scope note:

Excluded	Why
Windows GUI real runtime (G3)	Program deferral; 3-4w scope
`bench/` (G8)	Upstream tooling, not a fork deliverable
Upstream-core focus-by-id / synchronous-scan APIs	FORK-ONLY binding; out of scope
Pytest 3.8/3.9 CVE lane	3.10+ lane already patched; deliberate
Phase D mux implementation	Blocked on Q1
Scaling, sunset, user-feedback-driven evolution, intent change	Operator-excluded

1.5 Lineage

Fork, not rewrite. Upstream is `ghostty-org/ghostty`. Fork point (merge base) = `b3dc3d794513c11633e7c3347935693957744e83`. Upstream head observed this run = `e3ebb1d1b` (2026-10-16). Fork is **307 ahead / 3 behind** upstream/main (observed: `git rev-list --left-right --count origin/main...upstream/main`). Fork delta vs merge base: **352 files, +74,959 / −233**. License is inherited from upstream (MIT) with the fork's own additions; no relicensing is claimed here (see §14).

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§2. Current usability

2.1 What a person can use today, by interface

#	Interface	Reachable?	Installed?	Exercised by a human?	Evidence
U1	Rust crate `khostty-vt`	crates.io `0.1.0`	yes (operator, 2026-09-18)	yes (install+test)	`v0.1.0` release; `docs/changelog/0.1.0.md`
U2	npm `@khostty/libghostty-vt-wasm`	npm `0.1.0`	yes	status UNKNOWN	npm published
U3	macOS `Khostty.app`	`Khostty-0.1.0-macos.zip`	yes	yes (keystroke round-trip 2026-09-20)	release asset; signed, not notarized
U4	Linux GTK `.deb`	`khostty_0.1.0_amd64.deb`	yes (WSL Fedora 44)	partially (Xvfb)	installer verified
U5	Windows `setup.exe`	`Khostty-0.1.0-windows-x86_64-setup.exe`	install/ uninstall verified	**no GUI window ever launched**	`+version` + ABI only
U6	CLI IPC (`+new-window` / `+new-tab` / `+toggle-quick-term`)	upstream	n/a	status UNKNOWN	3 actions only
U7	Agent IPC (`khostty.sock` , 12-13 commands)	**NO — server never bound**	n/a	**no**	§7, §16

2.2 The nine-state ladder (per interface, worst case governs)

Interface	Intended	Implemented	Built	Distributed	Installed	Reachable	Exercised	Useful
Rust crate	✅	✅	✅	✅	✅	✅	✅	✅
WASM	✅	✅	✅	✅	⚠️ unknown	✅	⚠️ unknown	✅
macOS app	✅	✅	✅	✅	✅	✅	✅	✅
Linux .deb	✅	✅	✅	✅	✅	✅	⚠️ partial	✅
Windows exe	✅	⚠️ scaffold	✅	✅	✅	⚠️ no window	❌	❌
Agent IPC	✅	✅	✅	n/a	n/a	❌	❌	❌

2.3 Representative E2E journeys

Journey	Status	Oracle
J1: `cargo add khostty-vt` → parse a byte stream	**PASS** (operator-run 2026-09-18)	parser output correctness
J2: launch `Khostty.app`, type a command, see output	**PASS** (operator-run 2026-09-20, keystroke round-trip)	interactive shell
J3: install `.deb` on Linux, launch GUI	**PARTIAL** (Xvfb only)	GUI window appears
J4: install Windows `setup.exe`, launch GUI window	**FAIL** (no window)	window appears
J5: boot khostty, `khostty-client ping` → `pong`	**NOT RUNNABLE** (no runtime binds the socket)	JSON `{"ok":true}`
J6: agent creates a pane, writes to it, reads state	**NOT RUNNABLE**	pane round-trip
J7: run the 84-case conformance corpus	**PASS** (84/84, 2026-09-17)	0 failures

Technical E2E proof vs user validation: J1/J2/J3/J7 are technical proofs run by the operator or the agent. **No journey in this list has been run by a person other than the operator.** There is no external adoption evidence. That is a fact of this snapshot, not a defect.

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§3. Source coverage

3.1 Repository state

Property	Value	Command
`main` HEAD	`8e86ec93a1fd0f0c18f39b5c2d9ad604c1ffbaca`	`git rev-parse main`
`main` subject	`docs(audit): map native terminal usability and finish gates (#34)`	`git log -1`
Upstream head	`e3ebb1d1b` (2026-10-16)	`git log -1 upstream/main`
Merge base	`b3dc3d794513c11633e7c3347935693957744e83`	`git merge-base`
Fork Δ	307 ahead / 3 behind	`git rev-list --left-right --count`
Fork delta size	352 files, +74,959 / –233	`git diff --shortstat <base> main`
Working tree	clean on `main`; 11 linked worktrees under `~/.jcode/scratch/`	`git worktree list`
`v0.1.0` tag	`7e1684f60e2ab03cca3e0b653c65b1b4e1247d10` (annotated → `202aad543`)	`git rev-parse v0.1.0`

3.2 Agent-IPC surface inventory (the differentiated subsystem)

File	Lines
`src/apprt/ipc/server.zig`	1,238
`src/apprt/ipc/pane.zig`	1,209
`src/apprt/ipc/protocol.zig`	1,129
`src/apprt/ipc/handler.zig`	1,003
`src/apprt/ipc/events.zig`	605
`src/apprt/ipc/app_host.zig`	574
`src/apprt/ipc/auth.zig`	526
`src/apprt/ipc/fake_host.zig`	446
`src/apprt/ipc/mod.zig`	252
`src/apprt/ipc/state.zig`	248
Total	**7,230**

3.3 Branch inventory (this snapshot)

Branch	Ahead of main	Action
`khostty-phase-b-gtk-spec-20261017` (#36)	7	open PR
`khostty-dependabot-config-20261017` (#39)	3	open PR
`khostty-repo-hygiene-20261017` (#40)	2	open PR
`khostty-phase-a-polish-20261017` (#41)	8	open PR
`khostty-handoff-20261017` (#43)	3	open PR
`khostty-phase-b-nongtk-20261017` (#44)	9	open PR
`test/pane-write-parser-race` (#46)	9	open PR, quarantined by #48
`khostty-fork-discipline-clean-20261017` (#47)	4	open PR
`chore/mergify-queue-setup` (#49)	—	open PR
`chore/prettier-sweep-20261017` (#50)	1	open PR (this run)
`lanes/d-ci-installer-20261016` (#30)	—	DRAFT PR
`session-20261017-pr-triage-stop`	1	pushed branch (session docs), no PR

3.4 Coverage honesty

- **Inspected by the agent this run:** `main` tree, `docs/audits-archive/2026-10-17-final/*`, the two mux-design session docs, all 11 open PRs' metadata, all 10 open issues' bodies, `AGENTS.md`, `docs/changelog/0.1.0.md` head, `src/apprt/windows/*` presence, prettier state across tracked files.
- **NOT inspected this run:** full Zig test run, full conformance run, the GTK runtime source, the macOS app source beyond its presence, actual release asset bytes, PyPI/Go registry state, any upstream subtree in depth. Claims about them are **inherited** from the committed audit-archive and are cited as such, not re-verified.

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§4. Architecture

4.1 The five cooperating planes

```
graph TB
  subgraph P1["Plane 1 — Terminal engine (inherited)"]
    VT["VT parser / state machine<br/>src/terminal/"]
    SURF["Surface.zig — render-ready surface"]
    REN["Renderer (Metal / OpenGL)"]
  end
  subgraph P2["Plane 2 — App runtimes"]
    EMB["embedded (macOS AppKit)"]
    GTK["gtk (Linux/FreeBSD)"]
    WIN["windows (SCAFFOLD)"]
    NONE["none (headless)"]
    WEB["browser (wasm32)"]
  end
  subgraph P3["Plane 3 — IPC (the fork's delta)"]
    SRV["ipc/server.zig"]
    HOST["ipc/app_host.zig"]
    AUTH["ipc/auth.zig"]
    PROTO["ipc/protocol.zig"]
  end
  subgraph P4["Plane 4 — Bindings & artifacts"]
    C["C ABI libghostty-vt"]
    RS["Rust khostty-vt"]
    GO["Go khostty-go"]
    PY["Python khostty_vt"]
    W["WASM npm pkg"]
  end
  subgraph P5["Plane 5 — Assurance"]
    CONF["conformance/ (84 cases)"]
    ZT["zig build test"]
    CI["GitHub Actions"]
  end
  VT --> SURF --> REN
  SURF --> EMB
  SURF --> GTK
  SURF --> WIN
  VT --> C
  C --> RS
  C --> GO
  C --> PY
  VT --> W
  SRV -. "NOT BOUND by any runtime" .-> HOST
  HOST --> SURF
  SRV --> AUTH
  SRV --> PROTO
  CONF --> C
  ZT --> VT
  CI --> ZT
```

The dotted edge is the fork's single unfinished link: `ipc/server.zig` is compiled and tested but **no runtime calls `Server.bind`**.

4.2 Agent-IPC data flow (as designed)

```
sequenceDiagram
  participant A as Agent / khostty-client
  participant S as ipc/server.zig
  participant H as ipc/app_host.zig
  participant P as ipc/pane.zig
  participant C as Surface.zig (core)
  A->>S: connect to khostty.sock
  A->>S: {"v":1,"id":N,"auth":"<token>","cmd":"pane.list"}
  S->>S: constant-time token check (auth.zig)
  S->>H: dispatch(cmd)
  H->>P: resolve pane registry (Manager -> Broker)
  P->>C: surface op (queueIo mailbox for pane.write)
  C-->>P: result
  P-->>H: result
  H-->>S: {"v":1,"id":N,"ok":true,"data":{...}}
  S-->>A: response
  Note over H,C: pane.write currently touches the parser<br/>cross-thread directly — mus
```

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§5. Goals tree

```
graph LR
  G0[G0 Repo & docs<br/>DONE] --> ROOT[Usable terminal or<br/>embedded terminal]
  G1[G1 Conformance<br/>DONE 84/84] --> ROOT
  G2[G2 C ABI<br/>DONE] --> ROOT
  CVP[CVP: library-first 0.1.0<br/>SHIPPED] --> ROOT
  G5[G5 Rust crate<br/>DONE] --> CVP
  G7[G7 WASM<br/>DONE] --> CVP
  MVP[MVP: agent-IPC runtime bound<br/>NOT DELIVERED] --> ROOT
  G4M[G4 IPC modules<br/>DONE 7230 LOC] --> MVP
  G4R[G4 IPC runtime bound<br/>NOT STARTED] --> MVP
  PB[Phase B B1-B6<br/>B3/B6 landed, B1/B2/B4/B5 blocked] --> G4R
  PC[Phase C Phase-1 design<br/>C1-C4 done] --> PD
  PD[Phase D Phase-1 impl<br/>BLOCKED on Q1] --> ROOT
  G3[G3 Windows GUI<br/>SCAFFOLD] --> ROOT
```

ID	Goal	State	Evidence
CVP	Library-first 0.1.0	**SHIPPED**	tag `7e1684f60`, release 2026-09-27
MVP	Agent-IPC runtime bound	**NOT DELIVERED**	no `Server.bind` call outside `ipc/`
G0	Repo & docs	DONE	14 hand-curated docs; #41 refreshes 5 stale claims
G1	Conformance	DONE	84/84, `b2c388c9b`
G2	C ABI	DONE	`libghostty-vt` .a/.dylib/.so
G4-modules	IPC surface	DONE	7,230 LOC, 458 tests, 12 commands
G4-runtime	IPC bound	**NOT STARTED**	§4.1 dotted edge
G5	Rust crate	DONE	crates.io 0.1.0
G6	Go binding	LOCAL	S7 closed (#27); not on pkg.go.dev
G7	WASM	DONE	npm 0.1.0; A6 bumps `wasm/package.json` in #41
G9	Docs	DONE	#41 + audit-archive
G10	v0.1.0 release	DONE	6 assets
G3	Windows runtime	SCAFFOLD	#30 draft; S0-S8 open
Phase B	G4 finish	PARTIAL	B3+B6 in #44; B1/B2/B4/B5 blocked
Phase C	Phase-1 design	PARTIAL	C1-C4 + Q1 matrix + risks landed
Phase D	Phase-1 impl	BLOCKED	Q1

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§6. Component / feature completeness matrix

Capability	Intended	Built	Exercised	Useful	Accepted
VT parsing (C ABI)	✅	✅	✅ 84/84	✅	⚠️
VT parsing (Rust)	✅	✅	✅	✅	⚠️
VT parsing (WASM)	✅	✅	⚠️	✅	⚠️
macOS GUI	✅	✅	✅	✅	⚠️ operator only
Linux GUI	✅	✅	⚠️ partial	✅	⚠️
Windows GUI	✅	⚠️ scaffold	❌	❌	❌
CLI IPC (3 actions)	✅	✅	⚠️	✅	⚠️
Agent IPC (12 cmds)	✅	✅	❌	❌	❌
Agent IPC (7 events)	✅	✅	❌	❌	❌
Conformance gate	✅	✅	✅	✅	✅ (operator)
Phase-1 mux shell	✅	⚠️ docs only	❌	❌	❌

§7. Mocks, stubs, immaturity

Directly observed this run and inherited from `docs/audits-archive/2026-10-17-final/stubs.md` (61 findings). The high-consequence items:

#	Location	Present behaviour	User consequence	Closure oracle
S-1	`src/apprt/ipc/app_host.zig` — `Server.bind` unreferenced from runtimes	socket never created	agent IPC unreachable	a booted runtime creates the socket; `ping` returns `{"ok":true}`
S-2	`src/apprt/ipc/app_host.zig:290–298`	`pane.write` touches the parser cross-thread, violating `protocol.md §7` limit 1	data race under load; PR #46's repro attempts this but its premise is wrong (#48)	`queueIo` mailbox hop lands; a correct repro passes
S-3	`src/apprt/windows/*`	all windowing + IPC return `error.Unimplemented`	Windows terminal cannot open a window	GUI window launches on Windows; S1-S6, S8 close
S-4	`src/apprt/windows/renderer.zig` (64 lines)	orphaned; 8 `Unimplemented`	none (dead code)	A3 deletes it (#41)
S-5	`src/apprt/windows/win32api.zig` (85 lines)	declares fn pointers, nothing invokes	none (dead code)	A7 deletes it (#41)
S-6	`app_host.zig:177–178, 208–232`	`pane.focus` / `pane.search` return `error.Unsupported`	agents cannot focus-by-id or scan synchronously	upstream-core API (out of FORK-ONLY scope) → disclose in docs
S-7	`pkg/opengl/egl.zig:470–473`	tautological `expectEqualStrings(X, X)` — a false control	a broken EGL arm would pass CI	A2 replaces it with a real assertion (#41)
S-8	`src/terminal/search/Thread.zig:197`	`.BADF` reachable → `unreachable` panic (ABRT)	test suite crashes on a latency race	A5 guards teardown (#41)
S-9	`wasm/package.json`	version `0.0.0` (published npm is 0.1.0)	version drift	A6 bumps it (#41)
S-10	5 doc claims in `SECURITY.md` /	understate the code		A1 refreshes them (#41)

#	Location	Present behaviour	User consequence	Closure oracle
	`PLATFORMS.md` / `protocol.md`		plans over-estimate remaining work	
S-11	`fake_host.zig` (446 lines)	test-only IPC backend	none — by design	n/a
S-12	Phase-1 mux	docs + Q1 matrix + risk doc only; no skeleton on `main`	Phase D unimplementable	C5 skeleton lands (in #44)

Not a stub despite appearances: derived `resourcesDir` returning `.{}` in `src/apprt/windows/mod.zig:7` is compile-time-gated for a runtime that is itself deferred (G3).

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§8. Testing

Suite	Command	Cases	Status (source)
Conformance	`conformance/build.sh`	84	**84/84 PASS** 2026-09-17 (`b2c388c9b`)
Zig unit	`zig build test`	3,925 collected	**3,899 pass / 25 skip / 1 pre-existing ABRT** (audit-archive `state.md` §3.1)
Agent-IPC	`zig build test-agent-ipc`	458	compiles + passes; **proves signatures, not runtime** (`main_agent_ipc.zig:26-35`)
Rust	`cd khostty-vt && cargo test`	—	inherited; not re-run this run
WASM	`cd wasm && npm test`	52	inherited
Python	`pytest` (162 tests)	162	CI job added PR #16
Fork discipline	`zig build check-fork-discipline`	—	guard; PR #47 adds advisory CI

Negative controls: the EGL test at `pkg/opengl/egl.zig:470-473` was a *false* control (S-7) — it always passed. That is the one known instrument defect; A2 fixes it.

Flake: `terminal.search.Thread.test_0` ABRTs via a libxev kqueue race, characterized in `docs/sessions/20261014-ghostty-native-mux/evidence/flake-investigation-20261015.md`; A5 adds a teardown guard (#41).

Gate failures: on `main`, 8 PR branches were failing `Lint & Format` purely on accumulated prettier drift (57 files). Cleared this run (§22).

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§9. Security

Aspect	State	Evidence
IPC auth	fail-closed, 32-byte hex token, constant-time SHA-256	`auth.zig`, `protocol.md §5`
Token delivery	`KHOSTTY_IPC_TOKEN` env or `KHOSTTY_IPC_TOKEN_FILE` (fallback `ipc.token` beside socket)	`auth.zig`
Socket paths	macOS `~/Library/Caches/khostty/ipc.sock`; Linux/BSD `$XDG_RUNTIME_DIR/khostty/ipc.sock`; Windows `%LOCALAPPDATA%\khostty\ipc.sock`	`auth.zig:188-198`
Windows pipe ACL	`PIPE_DACL_SDDL = "D:P(A;;GA;;;OW)"`, `PIPE_HANDLE_INHERITABLE = false`	`windows/ipc.zig`
Command auth	every command except `ping` requires the token	`protocol.md`
Rate limiting	**absent** (B3 adds `max_connections` + per-conn limit; landed in #44)	stubs MVP-LAZY #9
Documented threat model	`docs/SECURITY.md` (partly stale until A1 lands)	#41
Secrets in tree	none observed; `SONAR_TOKEN` still an operator-pending secret	audit-archive

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§10. Privacy

Aspect	State
Data sent off-device by the terminal	None observed. The parser and IPC are local.
Telemetry	None observed in the fork (no analytics calls found in the inspected tree).
Logs	`ipc/events.zig` publishes local events; no remote sink observed.
PII in the repo	None observed in the inspected paths.
Compliance claim	**None made.** No privacy certification is claimed.

This section is deliberately thin because the evidence is thin: **the fork has no observed data-egress path.** If any is added, this section must be updated before that ships.

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§11. Integrity

Artifact	Signed	Notarized	Hash-verified	Rollback path
macOS `Khostty-0.1.0-macos.zip`	✅	❌	✅ (`shasum -c` 9/9)	reinstall prior `.app`
Linux `.deb`	n/a	n/a	✅	`dpkg -r`
Windows `setup.exe`	⚠️	n/a	✅	uninstaller verified
crates.io `khostty-vt`	n/a	n/a	provenance via crates.io	`cargo update --precise`
npm WASM	n/a	n/a	provenance via npm	version pin

Notarization is the one integrity gap on the primary desktop artifact. It matters for a downloaded `.app` on a non-developer machine.

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§12. UX / AX / DX

Dimension	State	Note
UX (end user)	macOS `.app` behaves like upstream; GTK partial; Windows no window	J2 passes, J3 partial, J4 fails
AX (accessibility)	inherited from upstream; not audited in the fork	UNKNOWN
DX (developer)	Read the CVPs: `cargo add` / `npm install` work	`khostty-vt/README.md`, `wasm/README.md`
Example programs	Go has `examples/`; Rust/WASM documented	inherited
Error messages	IPC returns structured JSON errors	`protocol.md`

No accessibility audit has been performed in the fork. That is an **unfinished obligation** if AX is in scope; it is listed in §21 but not silently deferred.

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§13. Research and alternatives

Comparator	Role	Status
Upstream Ghostty (pinned)	the inheritance source	tracked; synced via #18
Current Ghostty	drift reference	3 commits ahead of fork
WezTerm / Kitty / native terminal	SOTA shortlist (seeded, not studied)	**not evaluated**
`alacritty`, `foot`, `tmux`	adjacent tools	**not evaluated**

Reuse direction taken: retain upstream terminal emulation + rendering; keep the patch queue narrow (IPC + the fork's docs/CI); reuse the upstream conformance approach. **Advantage hypothesis:** "a specific embedding/workflow improvement with no unacceptable terminal correctness regression" — the agent-IPC surface is that improvement, and it is **not yet evaluated** because it is not yet runnable (MVP gap). This is honestly stated, not claimed as achieved.

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§14. Fork / rewrite state

Aspect	State
Type	**Fork** (not rewrite)
Upstream	`ghostty-org/ghostty`
Merge base	`b3dc3d794513c11633e7c3347935693957744e83`
Divergence	307 ahead / 3 behind (upstream `e3ebb1d1b`)
Fork delta	352 files, +74,959 / −233
Retained unchanged	terminal engine, renderers, macOS/GTK runtimes, CLI IPC
Intentionally changed	agent-IPC subsystem, fork docs, CI workflows, Windows scaffold
License	upstream MIT inherited; fork additions under the same terms (no relicensing claimed)
Update path	`git fetch upstream` + sync PR (last: #18, 397 commits)
Data migration	n/a (no persisted user data format owned by the fork)
Fork acceptance	the fork's own conformance corpus + `check-fork-discipline` guard
Protected paths	`src/terminal/`, `src/Surface.zig`, `src/Command.zig`, `src/runner.zig` — require a `// FORK:` block + an open fork issue

§15. Dependencies

Dependency	Kind	Version pin	Last upstream check
Zig	build	0.16.0 (CI)	n/a
libxev	runtime (async)	vendored upstream	via upstream sync
Metal / OpenGL / EGL	runtime (render)	system	n/a
Rust toolchain	build (crate)	inherited	n/a
Node/npm	build (WASM)	inherited	n/a
Go	build (binding)	inherited	S7 closed
Python 3.10+	build (binding)	3.10+ lane patched	3.8/3.9 deliberately unfixed
GitHub Actions runners	CI	standard Linux/macOS	n/a
Infisical	CI secret sync	`Infisical Sync` workflow	n/a

---

§16. Critical path

```
graph LR
  Q1{{Q1: pick Phase-1 runtime}} --> PD[Phase D: Phase-1 impl]
  HOST{{GTK-on-Linux host}} --> B1[B1: bind Server from GTK]
  B1 --> B2[B2: pane.write queueIo hop]
  B1 --> B4[B4: event publish hooks]
  B1 --> B5[B5: snapshot counters]
  B2 --> MVP[MVP: agent-IPC reachable]
  B4 --> MVP
  B5 --> MVP
  S1[S1-S6: Windows GUI] --> WG[Windows window gate]
  WG --> G3[G3 Windows runtime]
  O1{{O1: PyPI trusted publisher}} --> PY[PyPI publish]
  O2{{O2: SONAR_TOKEN}} --> SON[SonarCloud]
```

Active blockers:

ID	Blocker	Consequence	Owner
B-GTK	no GTK-on-Linux host reachable (Tailscale desk host offline)	B1/B2/B4/B5 cannot be built or verified	operator
B-Q1	Phase-1 runtime decision unmade	Phase D cannot start	operator
B-WIN	no Windows GUI launch rig in use	G3 window gate open	operator
B-DISK	build host earlier at 96-98% disk	release builds may fail	operator
B-SECRET	`SONAR_TOKEN`, PyPI trusted publisher	two CI/release hooks pending	operator

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§17. Operating cost

Cost centre	Observation
CI wall time	Trunk Check ~9-10 min; Test job long-running; several workflows cancelled/queued repeatedly
CI noise	`Vouch – Manage by Is…` re-runs in a cancel loop on `main` (observed 6+ cancelled runs in 90s)
Build host disk	earlier 96-98% full; release build failed on disk, debug build succeeded in 16m33s
Dependency maintenance	3 commits behind upstream; sync cadence via PR #18
Formatting debt	57 tracked files drifting → every doc-touching PR red (cleared this run)

---

§18. Decisions

Open

ID	Decision	Options	Owner	Status
Q1	Phase-1 windowed runtime	(a) Linux+GTK / (b) embedded-on-macOS / (c) mock-only	operator	**open**; mux-design recommends (c)
Q2	GTK runtime commitment	commit to GTK as Phase-1 target / defer	operator	open
Q3	Dependency policy	pin / track upstream head	operator	open
Q4	Windows GUI timing	now / after MVP	operator	open
—	PyPI publish (O1)	publish `khostty_vt`	operator	pending
—	SonarCloud token + App uninstall (O2)	set secret / uninstall App	operator	pending
—	Branch protection (O3/O4)	enable	operator (admin)	pending
—	PR #46 fate	fix repro / close	operator	quarantined by #48

Closed (this arc)

ID	Decision	Date	Rationale
D1	Phase-0 build wiring merged	2026-10-15	PR #15 / `ebc033eba`
D2	Fork-only; no upstream publishing	2026-10-15	operator binding; memory `mem_1792099237411_…`
D3	Phase A polish batch authored	2026-10-17	PR #41
D4	Prettier drift cleared on 8 branches + sweep PR	2026-10-17	this run

§19. Risks

Rank	Risk	Severity	Mitigation	Owner
R1	Agent-IPC runtime never wired	High	B1 on a GTK host; or pick option (c) and ship the skeleton honestly	operator + agent
R2	Q1 unmade	High	operator picks (c) per mux-design	operator
R3	Windows GUI never launched	Medium	#30 + S8 verification	operator
R4	`pane.write` race ships unfixed	High	B2 with B1; #48 corrects the repro	agent
R5	Prettier/format drift recurs	Low	sweep PR #50; add a scheduled full-repo pass	agent
R6	Build host disk exhaustion	Medium	free ≥50 GiB before release builds	operator
R7	Upstream drift grows	Low	sync cadence (3 behind now)	agent
R8	Accessibility never audited	Medium	add to §21; not deferred silently	agent
R9	`pane.focus` / `pane.search` unsupported	Medium	disclose in docs (A1)	agent
R10	PRs stall at red CI	Low	cleared this run	agent

---

§20. Estimates

Estimates are low / likely / high, calibrated to this fork's own observed rates (e.g., PR #15 Phase-0 build wiring: ~6d from branch to merged; A1-A7 batch: ~3h authored in one run; prettier sweep: ~1h this run). No P50/P90 is claimed.

Horizon	Deliverable	Low	Likely	High	Critical-path dependency
30 min	merge the 8 prettier-cleared branches (CI green)	0.5h	1h	2h	operator merge
1 h	merge PR #41 (Phase A) + #44 (B3/B6) + #50 (sweep)	1h	1.5h	3h	operator merge
4 h	hand-resolve #48, re-author a correct `pane.write` repro	3h	4h	6h	agent
24 h	re-run conformance + full Zig suite on the merged tree; publish the verified green	20h	24h	30h	build host
CVP	**already shipped** (library-first 0.1.0)	—	done	—	—
MVP	agent-IPC socket reachable on ≥1 runtime	1d	2d	4d	GTK host + Q1
Mature	CVP+MVP + Windows GUI + Phase-1 mux shell	2w	4w	8w	Q1 + Windows rig + Phase D

Aggregate agent effort vs critical path: the agent can author ~all remaining *code/docs* in **days**, but the critical path is gated by **operator-provided hosts and one decision (Q1)**. Those waits dominate; agent throughput is not the constraint.

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§21. Next actions

ID	Action	Depends on	Acceptance	Owner
N1	Merge the 8 prettier-cleared branches once CI green	—	CI green	operator
N2	Merge #41 (Phase A: A1-A7)	—	docs+code polish on main	operator
N3	Merge #44 (B3+B6+C1-C4)	—	hardening + design on main	operator
N4	Decide #48 vs #46	—	repro corrected or #46 closed	operator
N5	Pick Q1 (recommend (c))	—	decision recorded	operator
N6	Provide/reach a GTK-on-Linux host	Q1	`zig build` green on GTK	operator
N7	B1: bind `Server` from GTK runtime	N6	socket appears; `ping` → `pong`	agent
N8	B2: `pane.write` `queueIo` hop	N7	correct repro passes	agent
N9	B4/B5: event hooks + snapshot counters	N7	events publish; counters read	agent
N10	Accessibility audit (AX)	—	identified gaps recorded	agent
N11	Disclose `pane.focus` / `pane.search` unsupported	—	`protocol.md` + `AGENT.md` updated (in #41)	agent
N12	Free ≥50 GiB on the build host	—	release build succeeds	operator
N13	Set `SONAR_TOKEN`; configure PyPI trusted publishing	—	hooks green	operator
N14	Notarize the macOS `.app`	Apple creds	`spctl` accepts downloaded app	operator
N15	S8: Windows rebuild + installer verification on post-sync tree	Windows rig	window gate or documented gap	operator+agent

§22. Evidence

Evidence	Value
`main` HEAD	`8e86ec93a1fd0f0c18f39b5c2d9ad604c1ffbaca`
`v0.1.0` tag	`7e1684f60e2ab03cca3e0b653c65b1b4e1247d10` → `202aad543`
Merge base w/ upstream	`b3dc3d794513c11633e7c3347935693957744e83`
Upstream head	`e3ebb1d1b`
Fork divergence	307 ahead / 3 behind
Fork delta	352 files, +74,959 / –233
Agent-IPC LOC	7,230 across 10 files
Conformance	84/84 @ `b2c388c9b`
Audit snapshot	`869b9c75e` (`docs/audits-archive/2026-10-17-final/`)
Session evidence	`b6977ebca` (STOP report + phenoreg-linker)
Prettier sweep PR	**#50** (this run)
Prettier per-branch commits	#36 `c53694b08` · #39 `cd642a279` · #40 `a58b2335f` · #41 `f1466ca47` · #43 `542962330` · #44 `71c5cd57b` · #46 `b88f0c600` + `07695a4ec` · #47 `650fef21b`
Open PRs	11 (#30, #36, #39, #40, #41, #43, #44, #46, #47, #49, #50)
Open issues	10 (#13, #20-#26, #28, #48); #27 closed
Observation time	2026-10-17T22:17Z

Confidence: high for anything cited by SHA and observed live this run; medium for inherited audit-archive test counts not re-run; UNKNOWN where marked.

§23. Closeout

What works today

1. `cargo add khostty-vt` → an instant VT parser in Rust.
2. `npm install @khostty/libghostty-vt-wasm` → an instant VT parser in the browser/Node.
3. A signed macOS `Khostty.app` that runs and takes keystrokes.
4. A Linux GTK `.deb` that installs (GUI launch verified only via Xvfb).
5. An 84/84 conformance gate over the C library.
6. A compiled, tested agent-IPC library (not yet runnable).

What changed this run

- Cleared accumulated prettier drift on **8 PR branches** and opened **PR #50** for `main`.
- Committed the audit-arc session evidence (`b6977ebca`).
- Published the **State & Forward Docket v1.0 spec** to PhenoRegistry.
- Published **this docket**.

What still does not work

- The agent-IPC socket is never created (MVP gap).
- `pane.write`'s cross-thread parser access (S-2).
- The Windows GUI never opens a window.
- `pane.focus` / `pane.search` return `error.Unsupported`.
- No accessibility audit exists.
- The macOS app is signed but not notarized.

Blockers and precise consequences

Blocker	Consequence
No GTK-on-Linux host	B1/B2/B4/B5 cannot be produced or verified → MVP cannot land
Q1 unmade	Phase D cannot start
No Windows GUI rig	G3 window gate stays open
Build-host disk	release builds may fail
Missing secrets (SONAR_TOKEN, Apple, PyPI)	three release/CI hooks stay pending

Next actual usable result

The next result a person can *use* is the already-shipped **library-first CVP**; the next result they can use *new* is the **merged Phase A polish** (docs stop understating the code). The next result that makes the fork *do what it was for* is **B1** — but only once a GTK-on-Linux host exists.

Assurance sign-off

- **Product owner sign-off:** pending (operator).
- **Assurance-owner sign-off:** pending (independent re-run of conformance + Zig suite on the merged tree).
- **Honesty statement:** no user session, feedback, adoption, subagent, commit, merge, or PDF-open is claimed here beyond what §22 records.

End of State & Forward Docket v1.0 — Khostty @ `8e86ec93a` *, 2026-10-17T22:17Z.*